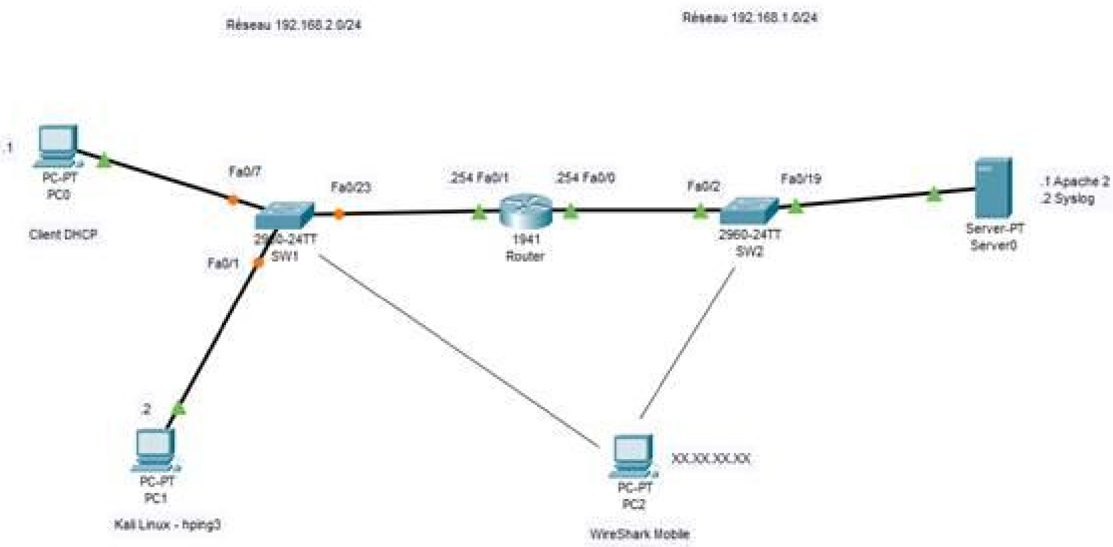
Compte Rendu TP - Sécurité Couche 4



Tests de ping

kali > routeur

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.2.254
PING 192.168.2.254 (192.168.2.254) 56(84) bytes of data.
64 bytes from 192.168.2.254: icmp_seq=1 ttl=255 time=0.963 ms
64 bytes from 192.168.2.254: icmp_seq=2 ttl=255 time=1.27 ms
64 bytes from 192.168.2.254: icmp_seq=3 ttl=255 time=1.14 ms
64 bytes from 192.168.2.254: icmp_seq=4 ttl=255 time=1.26 ms
^C
--- 192.168.2.254 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3049ms
rtt min/avg/max/mdev = 0.963/1.157/1.265/0.123 ms
```

syslog > routeur

```
PS C:\Users\rbuchenaud> ping 192.168.1.254

Envoi d’une requête 'Ping'  192.168.1.254 avec 32 octets de données :
Réponse de 192.168.1.254 : octets=32 temps<1ms TTL=255
Réponse de 192.168.1.254 : octets=32 temps<1ms TTL=255
Réponse de 192.168.1.254 : octets=32 temps<1ms TTL=255
Réponse de 192.168.1.254 : octets=32 temps<1ms TTL=255

Statistiques Ping pour 192.168.1.254:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
PS C:\Users\rbuchenaud>
```

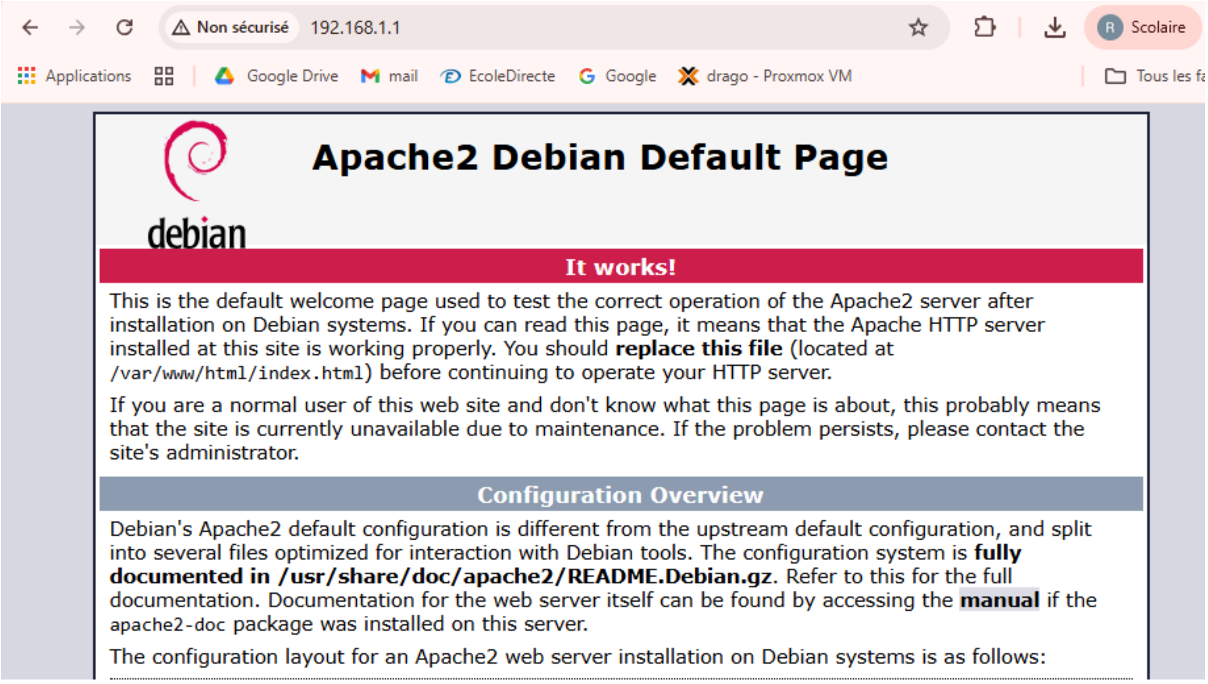
syslog > serveur web

```
PS C:\Users\rbuchenaud> ping 192.168.1.1

Envoi d’une requête 'Ping'  192.168.1.1 avec 32 octets de données :
Réponse de 192.168.1.1 : octets=32 temps<1ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps<1ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps<1ms TTL=64
Réponse de 192.168.1.1 : octets=32 temps<1ms TTL=64

Statistiques Ping pour 192.168.1.1:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms
PS C:\Users\rbuchenaud>
```

Serveur HTTP Fonctionnel



Apache2 Debian Default Page

debian

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

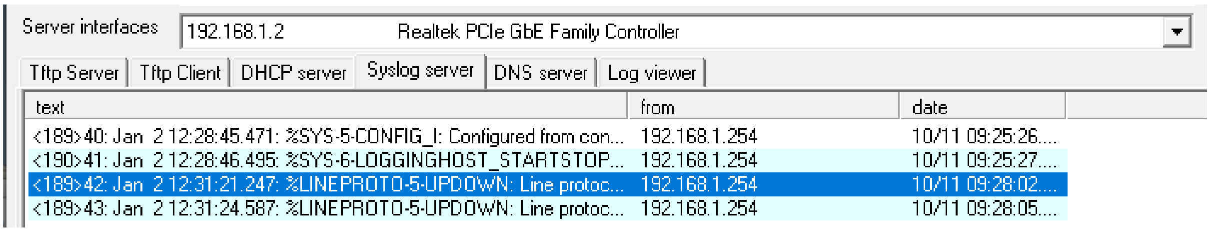
Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

Mise en place de l’autorisation du serveur syslog :

```
Router(config)#logging host 192.168.1.2
Router(config)#exit
Router#
Jan  2 12:28:45.471: %SYS-5-CONFIG_I: Configured from console by console
Jan  2 12:28:46.495: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.2 p
ort 514 started - CLI initiated
```

connexion du routeur sur syslog pour tester :



text	from	date
<189>40: Jan 2 12:28:45.471: %SYS-5-CONFIG_I: Configured from con...	192.168.1.254	10/11 09:25:26....
<190>41: Jan 2 12:28:46.495: %SYS-6-LOGGINGHOST_STARTSTOP...	192.168.1.254	10/11 09:25:27....
<189>42: Jan 2 12:31:21.247: %LINEPROTO-5-UPDOWN: Line protoc...	192.168.1.254	10/11 09:28:02....
<189>43: Jan 2 12:31:24.587: %LINEPROTO-5-UPDOWN: Line protoc...	192.168.1.254	10/11 09:28:05....

Port-mirroring SW1

Source : router gi0/23
Destination : wireshark gi0/11

```
SW1(config)#monitor session 1 source int gi0/23
SW1(config)#monitor session 1 source int gi0/11
```

Port-mirroring SW2

```
Switch(config)#monitor session 1 source interface gigabitEthernet 0/2
Switch(config)#monitor session 1 destination interface gigabitEthernet 0/11
```

Source : Routeur Gi0/23

Destination : WireShark Mobile Gi011

Vérification du fonctionnement du Port Mirroring sur chacun des routeurs :

SW1

No.	Time	Source	Destination	Protocol	Lengtl	Info
882	85.382747	192.168.2.1	172.17.10.1	DNS	75	Standard query 0xf10c HTTPS docs.google.com
883	85.383416	192.168.2.1	172.17.10.1	DNS	75	Standard query 0xcaad A docs.google.com
884	85.383630	192.168.2.254	192.168.2.1	ICMP	70	Destination unreachable (Host unreachable)
885	85.523073	192.168.2.1	172.17.10.1	DNS	75	Standard query 0x1062 A wpad.info.local
886	85.523308	192.168.2.1	172.17.10.2	DNS	75	Standard query 0x1062 A wpad.info.local
887	85.828861	GigaByteTech_e6:4b:…	Broadcast	ARP	60	Who has 172.17.10.2? Tell 172.17.13.15
888	85.946205	GigaByteTech_e6:4b:…	Broadcast	ARP	60	Who has 172.17.10.1? Tell 172.17.13.15
889	85.956203	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.0.254? Tell 172.17.13.13
890	85.956215	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.2? Tell 172.17.13.13
891	86.335514	192.168.2.1	172.17.10.1	DNS	77	Standard query 0xfd2e A beacons3.gvt2.com
892	86.335761	192.168.2.1	172.17.10.2	DNS	77	Standard query 0xfd2e A beacons3.gvt2.com
893	86.336318	192.168.2.254	192.168.2.1	ICMP	70	Destination unreachable (Host unreachable)
894	86.383231	192.168.2.1	172.17.10.1	DNS	76	Standard query 0x29e8 A dns.msftncsi.com
895	86.386224	192.168.2.1	23.48.32.82	TCP	66	52032 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
896	86.398367	192.168.2.1	172.17.10.2	DNS	75	Standard query 0xb042 HTTPS docs.google.com
897	86.398724	192.168.2.1	172.17.10.2	DNS	75	Standard query 0x778a A docs.google.com
898	86.413755	192.168.2.1	172.17.10.2	DNS	76	Standard query 0x29e8 A dns.msftncsi.com
899	86.446858	GigaByteTech_e6:4b:…	Broadcast	ARP	60	Who has 172.17.10.2? Tell 172.17.13.15
900	86.523229	192.168.2.1	172.17.10.1	DNS	89	Standard query 0x9bdc HTTPS clientservices.googleapis.com
901	86.523565	192.168.2.1	172.17.10.2	DNS	89	Standard query 0xbb6d A clientservices.googleapis.com
902	86.664797	192.168.2.1	172.17.10.1	DNS	76	Standard query 0x2579 A beacons.gvt2.com
903	86.694979	192.168.2.1	172.17.10.2	DNS	76	Standard query 0x2579 A beacons.gvt2.com
904	86.851074	192.168.2.1	172.17.10.1	DNS	97	Standard query 0xa82f A array817.prod.do.dsp.mp.microsoft.com
905	86.851313	192.168.2.1	172.17.10.2	DNS	97	Standard query 0xa82f A array817.prod.do.dsp.mp.microsoft.com
126	8.516791	192.168.1.2	172.17.10.1	DNS	77	Standard query 0xa9b5 HTTPS beacons4.gvt2.com
127	8.517155	192.168.1.2	172.17.10.1	DNS	77	Standard query 0x9c45 A beacons4.gvt2.com
128	8.517638	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)
129	8.543536	192.168.1.2	172.17.10.1	DNS	79	Standard query 0x9d59 A accounts.google.com
130	8.569800	192.168.1.2	172.17.10.2	DNS	79	Standard query 0x9d59 A accounts.google.com
131	8.629366	192.168.1.2	172.17.10.1	DNS	75	Standard query 0x32d4 A ssl.gstatic.com
132	8.655124	192.168.1.2	172.17.10.2	DNS	75	Standard query 0x32d4 A ssl.gstatic.com
133	8.741860	192.168.1.2	172.17.10.2	DNS	81	Standard query 0x5221 HTTPS waa-pa.googleapis.com
134	8.742227	192.168.1.2	172.17.10.2	DNS	81	Standard query 0xd71a A waa-pa.googleapis.com
135	9.065636	192.168.1.2	172.17.10.1	DNS	79	Standard query 0x4485 HTTPS clients4.google.com

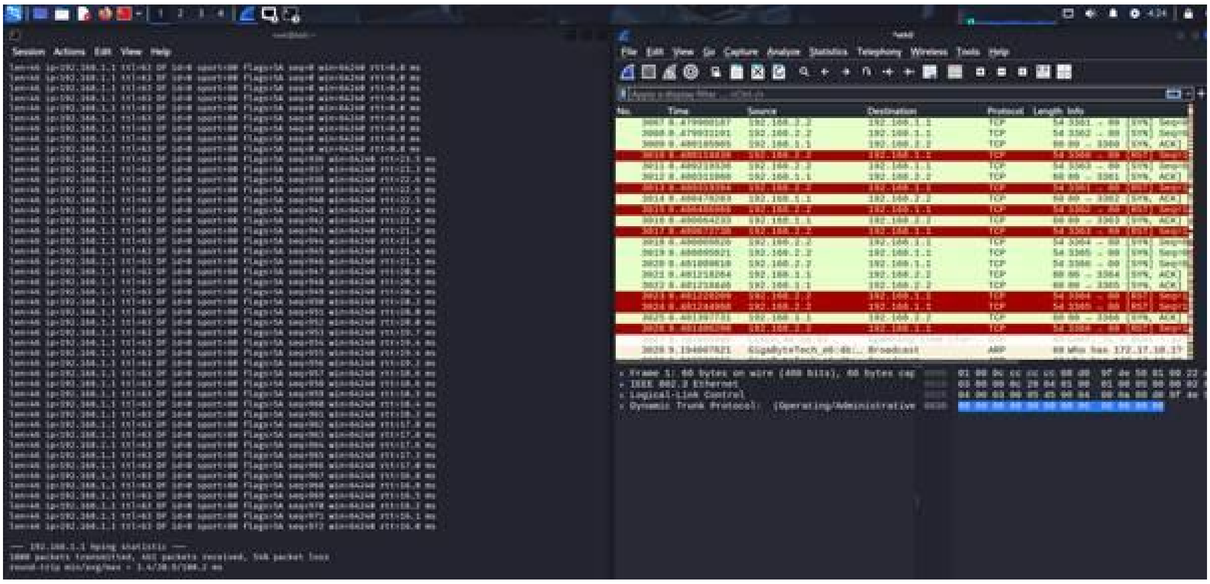
Fin de l'installation et de la vérification de son fonctionnement.

Attaques avant application des contre-mesures :

Le serveur est en effet ralentit par les attaques, les pings depuis un client sont plus longs ou n’arrivent tout simplement pas :

```
Statistiques Ping pour 192.168.1.1:
    Paquets : envoyés = 129, reçus = 125, perdus = 4 (perte 3%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 29ms, Moyenne = 0ms
```

Capture hping3 WireShark sur Kali



10 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
2203	265.287554	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1795 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2204	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1794 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2205	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1793 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2206	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1792 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2207	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1791 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2208	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1790 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2209	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1789 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2210	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1788 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2211	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1787 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2212	265.288414	192.168.2.2	192.168.1.1	TCP	60	1795 → 80 [RST] Seq=1 Win=0 Len=0
2213	265.288631	192.168.2.2	192.168.1.1	TCP	60	1794 → 80 [RST] Seq=1 Win=0 Len=0
2214	265.288631	192.168.2.2	192.168.1.1	TCP	60	1793 → 80 [RST] Seq=1 Win=0 Len=0
2215	265.288631	192.168.2.2	192.168.1.1	TCP	60	1792 → 80 [RST] Seq=1 Win=0 Len=0
2216	265.288631	192.168.2.2	192.168.1.1	TCP	60	1791 → 80 [RST] Seq=1 Win=0 Len=0
2217	265.289105	192.168.2.2	192.168.1.1	TCP	60	1790 → 80 [RST] Seq=1 Win=0 Len=0
2218	265.289318	192.168.2.2	192.168.1.1	TCP	60	1789 → 80 [RST] Seq=1 Win=0 Len=0
2219	265.289318	192.168.2.2	192.168.1.1	TCP	60	1788 → 80 [RST] Seq=1 Win=0 Len=0
2220	265.289318	192.168.2.2	192.168.1.1	TCP	60	1787 → 80 [RST] Seq=1 Win=0 Len=0

100 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
301	5.356533	192.168.1.1	192.168.2.2	TCP	60	80 → 1687 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
302	5.357058	192.168.1.1	192.168.2.2	TCP	60	80 → 1688 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
303	5.357269	192.168.1.1	192.168.2.2	TCP	60	80 → 1689 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
304	5.357517	192.168.1.1	192.168.2.2	TCP	60	80 → 1690 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
305	5.357728	192.168.1.1	192.168.2.2	TCP	60	80 → 1691 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
306	5.357728	192.168.1.1	192.168.2.2	TCP	60	80 → 1692 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
307	5.359455	192.168.2.2	192.168.1.1	TCP	60	1684 → 80 [RST] Seq=1 Win=0 Len=0
308	5.359455	192.168.2.2	192.168.1.1	TCP	60	1685 → 80 [RST] Seq=1 Win=0 Len=0
309	5.359455	192.168.2.2	192.168.1.1	TCP	60	1686 → 80 [RST] Seq=1 Win=0 Len=0
310	5.359696	192.168.2.2	192.168.1.1	TCP	60	1687 → 80 [RST] Seq=1 Win=0 Len=0
311	5.359696	192.168.2.2	192.168.1.1	TCP	60	1688 → 80 [RST] Seq=1 Win=0 Len=0
312	5.359696	192.168.2.2	192.168.1.1	TCP	60	1689 → 80 [RST] Seq=1 Win=0 Len=0
313	5.359696	192.168.2.2	192.168.1.1	TCP	60	1690 → 80 [RST] Seq=1 Win=0 Len=0
314	5.359696	192.168.2.2	192.168.1.1	TCP	60	1691 → 80 [RST] Seq=1 Win=0 Len=0
315	5.359696	192.168.2.2	192.168.1.1	TCP	60	1692 → 80 [RST] Seq=1 Win=0 Len=0
316	5.359696	192.168.2.2	192.168.1.1	TCP	60	1693 → 80 [SYN] Seq=0 Win=512 Len=0
317	5.359696	192.168.2.2	192.168.1.1	TCP	60	1694 → 80 [SYN] Seq=0 Win=512 Len=0
318	5.359952	192.168.2.2	192.168.1.1	TCP	60	1695 → 80 [SYN] Seq=0 Win=512 Len=0
319	5.359952	192.168.2.2	192.168.1.1	TCP	60	1696 → 80 [SYN] Seq=0 Win=512 Len=0

1000 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
1866	15.578846	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1911 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1867	15.579058	192.168.2.2	192.168.1.1	TCP	60	1910 → 80 [RST] Seq=1 Win=0 Len=0
1868	15.579058	192.168.2.2	192.168.1.1	TCP	60	1911 → 80 [RST] Seq=1 Win=0 Len=0
1869	15.579413	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1912 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1870	15.579623	192.168.2.2	192.168.1.1	TCP	60	1912 → 80 [RST] Seq=1 Win=0 Len=0
1871	15.579833	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1913 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1872	15.580043	192.168.2.2	192.168.1.1	TCP	60	1913 → 80 [RST] Seq=1 Win=0 Len=0
1873	15.582808	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1914 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1874	15.583021	192.168.2.2	192.168.1.1	TCP	60	1914 → 80 [RST] Seq=1 Win=0 Len=0
1875	15.583021	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1915 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1876	15.583393	192.168.2.2	192.168.1.1	TCP	60	1915 → 80 [RST] Seq=1 Win=0 Len=0
1877	15.583701	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1916 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1878	15.583914	192.168.2.2	192.168.1.1	TCP	60	1916 → 80 [RST] Seq=1 Win=0 Len=0
1879	15.583914	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1917 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1880	15.584322	192.168.2.2	192.168.1.1	TCP	60	1917 → 80 [RST] Seq=1 Win=0 Len=0
1881	15.586633	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1918 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1882	15.586847	192.168.2.2	192.168.1.1	TCP	60	1918 → 80 [RST] Seq=1 Win=0 Len=0
1883	15.586847	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1919 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1884	15.587255	192.168.2.2	192.168.1.1	TCP	60	1919 → 80 [RST] Seq=1 Win=0 Len=0
1885	15.587518	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1920 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1886	15.587728	192.168.2.2	192.168.1.1	TCP	60	1920 → 80 [RST] Seq=1 Win=0 Len=0
1887	15.587937	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1921 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1888	15.588147	192.168.2.2	192.168.1.1	TCP	60	1921 → 80 [RST] Seq=1 Win=0 Len=0
1889	15.590715	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1922 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1890	15.590927	192.168.2.2	192.168.1.1	TCP	60	1922 → 80 [RST] Seq=1 Win=0 Len=0

flood :

No.	Time	Source	Destination	Protocol	Lengtl	Info
4102…	232.961261	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62016 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961261	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62015 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961517	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62014 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961517	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62013 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961517	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62011 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961517	192.168.2.2	192.168.1.1	TCP	60	62031 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.961517	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62082 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961517	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62144 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961760	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62112 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961760	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62055 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961760	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62116 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961760	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62071 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961760	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62025 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961990	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62018 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961990	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62106 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.961990	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 62087 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62030 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62029 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62028 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62027 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62026 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62024 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62023 → 80 [RST] Seq=1 Win=0 Len=0
4102…	232.963537	192.168.2.2	192.168.1.1	TCP	60	62022 → 80 [RST] Seq=1 Win=0 Len=0

Les trames en rouge et noir détectent une réutilisation de port par l'attaque qui ne devrait pas être.

Suite à l'attaque, mise en place des contre-mesures sur le routeur :

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip access-list extended TCP_INTER
Router(config-ext-nacl)#10 permit tcp any host 192.168.1.1 eq www
Router(config-ext-nacl)#exit
Router(config)#ip tcp intercept mode intercept
Router(config)#ip tcp intercept list TCP_INTER
Router(config)#ip tcp intercept one-minute low 50
```

```
Router(config)#ip tcp intercept one-minute low 50 high 100
```

Avant les nouvelles attaques : les connexions avant

```
Router#sh
Jan  2 13:39:23.343: %SYS-5-CONFIG_I: Configured from console by consoletcp intercept connections
Incomplete:
Client                 Server                 State    Create   Timeout  Mode

Established:
Client                 Server                 State    Create   Timeout  Mode
```

Après les nouvelles attaques : beaucoup de nouvelles connexion incomplètes.

10

```
Router#sh tcp intercept connections
Incomplete:
Client                 Server                 State    Create   Timeout  Mode
192.168.2.2:2259       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2258       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2257       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2256       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2263       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2262       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2261       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2260       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2265       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I
192.168.2.2:2264       192.168.1.1:80         SYNRCVD  00:00:09 00:00:05 I

Established:
Client                 Server                 State    Create   Timeout  Mode
```

100

```
Router#sh tcp intercept connections
Incomplete:
Client                 Server                 State    Create   Timeout  Mode
192.168.2.2:1299       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1298       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1297       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1296       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1283       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1282       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1281       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1280       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1287       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1286       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1285       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1284       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1291       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1290       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1289       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1288       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1295       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1294       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1293       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
192.168.2.2:1292       192.168.1.1:80         SYNRCVD  00:00:22 00:00:08 I
```

```
Jan  2 13:48:08.331: %TCP-6-INTERCEPT: getting aggressive, count (1100/1100) 1 min 0
```

```
Router#sh tcp intercept stat
Intercepting new connections using access-list TCP_INTE
731 incomplete, 0 established connections (total 731)
3950 connection requests per minute
```

flood

```
Router#
Jan  2 13:50:17.527: %TCP-6-INTERCEPT: calming down, count (0/900) 1 min 0
Jan  2 13:50:25.243: %TCP-6-INTERCEPT: getting aggressive, count (1100/1100) 1 min 0
Jan  2 13:50:45.915: %TCP-6-INTERCEPT: calming down, count (899/900) 1 min 0
```

Le routeur détecte l'excès de 100 trames SYN par minute et passe en “getting aggresive”. Les nouvelles trames sont bloquées. Si elle en détecte ensuite moins de 50 trames par minute, il les laisse à nouveau passer.

```
Router#sh tcp intercept stat
Intercepting new connections using access-list TCP_INTER
0 incomplete, 0 established connections (total 0)
3733 connection requests per minute
```

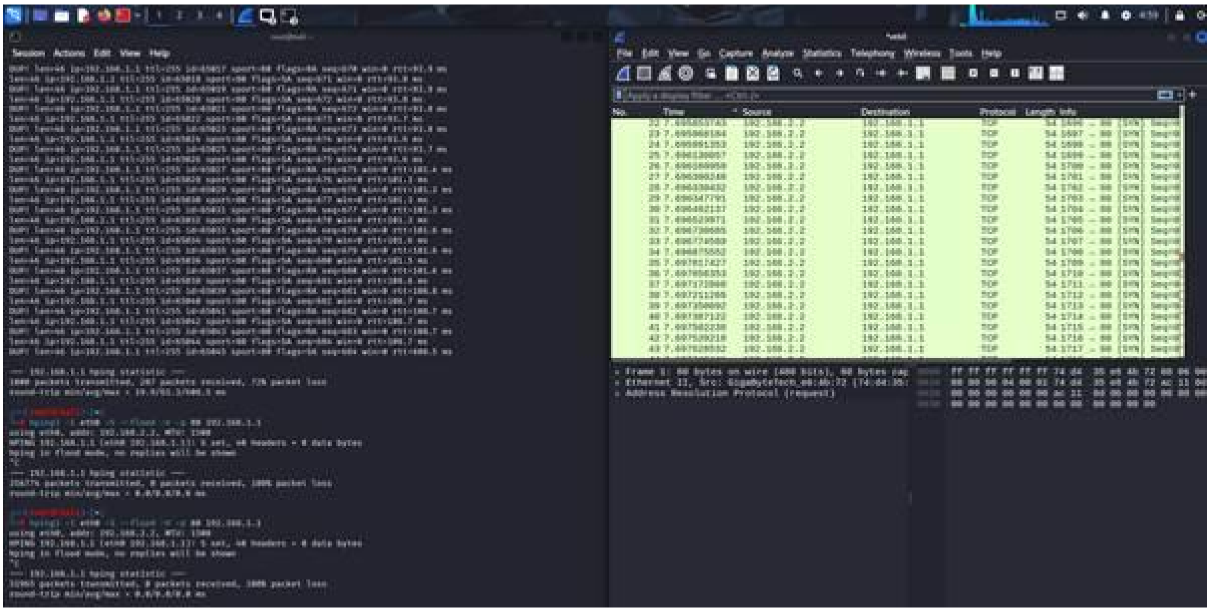
Toutes les connexions incomplètes ne sont pas présentes car elles sont bloquées à partir d'un certain nombre.

Une fois les contre-mesures appliquées, les paquets sont bloqués. On le voit sur les statistiques de hping3 :

```
┌──(root㉿kali)-[~]
└─# hping3 -I eth0 -S --flood -V -p 80 192.168.1.1
using eth0, addr: 192.168.2.2, MTU: 1500
HPING 192.168.1.1 (eth0 192.168.1.1): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.1.1 hping statistic ---
139469 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
┌──(root㉿kali)-[~]
└─# hping3 -I eth0 -i u1 -S -p 80 -c 100 192.168.1.1
HPING 192.168.1.1 (eth0 192.168.1.1): S set, 40 headers + 0 data bytes

--- 192.168.1.1 hping statistic ---
100 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```



Les paquets sont bien envoyés, mais bloqués par le routeur. Il a détecté l'aggression

Enfin, on utilise le port mirroring afin de vérifier que les paquets sont bloqués par le routeur dès que c'est nécessaire.

Pour le réseau 192.168.2.0, toutes les trames passent peu importe leur nombre :

Pour 10 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
2203	265.287554	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1795 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2204	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1794 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2205	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1793 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2206	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1792 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2207	265.287797	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1791 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2208	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1790 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2209	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1789 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2210	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1788 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2211	265.288031	192.168.1.1	192.168.2.2	TCP	60	[TCP Retransmission] 80 → 1787 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2212	265.288414	192.168.2.2	192.168.1.1	TCP	60	1795 → 80 [RST] Seq=1 Win=0 Len=0
2213	265.288631	192.168.2.2	192.168.1.1	TCP	60	1794 → 80 [RST] Seq=1 Win=0 Len=0
2214	265.288631	192.168.2.2	192.168.1.1	TCP	60	1793 → 80 [RST] Seq=1 Win=0 Len=0
2215	265.288631	192.168.2.2	192.168.1.1	TCP	60	1792 → 80 [RST] Seq=1 Win=0 Len=0
2216	265.288631	192.168.2.2	192.168.1.1	TCP	60	1791 → 80 [RST] Seq=1 Win=0 Len=0
2217	265.289105	192.168.2.2	192.168.1.1	TCP	60	1790 → 80 [RST] Seq=1 Win=0 Len=0
2218	265.289318	192.168.2.2	192.168.1.1	TCP	60	1789 → 80 [RST] Seq=1 Win=0 Len=0
2219	265.289318	192.168.2.2	192.168.1.1	TCP	60	1788 → 80 [RST] Seq=1 Win=0 Len=0
2220	265.289318	192.168.2.2	192.168.1.1	TCP	60	1787 → 80 [RST] Seq=1 Win=0 Len=0

Pour 1000 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
1866	15.578846	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1911 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1867	15.579058	192.168.2.2	192.168.1.1	TCP	60	1910 → 80 [RST] Seq=1 Win=0 Len=0
1868	15.579058	192.168.2.2	192.168.1.1	TCP	60	1911 → 80 [RST] Seq=1 Win=0 Len=0
1869	15.579413	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1912 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1870	15.579623	192.168.2.2	192.168.1.1	TCP	60	1912 → 80 [RST] Seq=1 Win=0 Len=0
1871	15.579833	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1913 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1872	15.580043	192.168.2.2	192.168.1.1	TCP	60	1913 → 80 [RST] Seq=1 Win=0 Len=0
1873	15.582808	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1914 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1874	15.583021	192.168.2.2	192.168.1.1	TCP	60	1914 → 80 [RST] Seq=1 Win=0 Len=0
1875	15.583021	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1915 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1876	15.583393	192.168.2.2	192.168.1.1	TCP	60	1915 → 80 [RST] Seq=1 Win=0 Len=0
1877	15.583701	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1916 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1878	15.583914	192.168.2.2	192.168.1.1	TCP	60	1916 → 80 [RST] Seq=1 Win=0 Len=0
1879	15.583914	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1917 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1880	15.584322	192.168.2.2	192.168.1.1	TCP	60	1917 → 80 [RST] Seq=1 Win=0 Len=0
1881	15.586633	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1918 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1882	15.586847	192.168.2.2	192.168.1.1	TCP	60	1918 → 80 [RST] Seq=1 Win=0 Len=0
1883	15.586847	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1919 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1884	15.587255	192.168.2.2	192.168.1.1	TCP	60	1919 → 80 [RST] Seq=1 Win=0 Len=0
1885	15.587518	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1920 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1886	15.587728	192.168.2.2	192.168.1.1	TCP	60	1920 → 80 [RST] Seq=1 Win=0 Len=0
1887	15.587937	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1921 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1888	15.588147	192.168.2.2	192.168.1.1	TCP	60	1921 → 80 [RST] Seq=1 Win=0 Len=0
1889	15.590715	192.168.1.1	192.168.2.2	TCP	60	[TCP Port numbers reused] 80 → 1922 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
1890	15.590927	192.168.2.2	192.168.1.1	TCP	60	1922 → 80 [RST] Seq=1 Win=0 Len=0

Pour le réseau 192.168.1.0, le routeur arrête les trames s'il en détecte plus de 100/minute :

Pour 1000 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
26	2.707245	192.168.1.2	172.17.10.1	DNS	96	Standard query 0x8b34 A disc801.prod.do.dsp.mp.microsoft.com
27	3.021719	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.1? Tell 172.17.13.13
28	3.021735	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.2? Tell 172.17.13.13
29	3.050342	192.168.1.2	172.17.10.1	DNS	75	Standard query 0x399d A ssl.gstatic.com
30	3.075375	192.168.1.2	172.17.10.1	DNS	75	Standard query 0x399d A ssl.gstatic.com
31	3.310445	192.168.1.2	172.17.10.1	DNS	86	Standard query 0xc746 HTTPS waa-pa.clients6.google.com
32	3.310679	192.168.1.2	172.17.10.1	DNS	86	Standard query 0x2dc8 A waa-pa.clients6.google.com
33	3.310885	192.168.1.2	172.17.10.1	DNS	75	Standard query 0xcf41 HTTPS play.google.com
34	3.311118	192.168.1.2	172.17.10.1	DNS	75	Standard query 0x2c03 A play.google.com
35	3.311382	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)
36	3.556561	192.168.2.2	192.168.1.1	TCP	60	1692 → 80 [RST] Seq=1 Win=0 Len=0
37	3.556831	192.168.2.2	192.168.1.1	TCP	60	1693 → 80 [RST] Seq=1 Win=0 Len=0
38	3.557101	192.168.2.2	192.168.1.1	TCP	60	1694 → 80 [RST] Seq=1 Win=0 Len=0
39	3.557326	192.168.2.2	192.168.1.1	TCP	60	1695 → 80 [RST] Seq=1 Win=0 Len=0
40	3.557584	192.168.2.2	192.168.1.1	TCP	60	1696 → 80 [RST] Seq=1 Win=0 Len=0
41	3.557785	192.168.2.2	192.168.1.1	TCP	60	1697 → 80 [RST] Seq=1 Win=0 Len=0
42	3.558054	192.168.2.2	192.168.1.1	TCP	60	1698 → 80 [RST] Seq=1 Win=0 Len=0
43	3.558255	192.168.2.2	192.168.1.1	TCP	60	1699 → 80 [RST] Seq=1 Win=0 Len=0
44	3.558523	192.168.2.2	192.168.1.1	TCP	60	1700 → 80 [RST] Seq=1 Win=0 Len=0

Pour 1000 trames :

No.	Time	Source	Destination	Protocol	Lengtl	Info
157	14.589133	192.168.1.2	172.17.10.1	DNS	75	Standard query 0xd3bc A play.google.com
158	14.589328	192.168.1.2	172.17.10.2	DNS	75	Standard query 0xd3bc A play.google.com
159	14.620598	192.168.1.2	172.17.10.2	DNS	75	Standard query 0xb599 HTTPS ssl.gstatic.com
160	14.620975	192.168.1.2	172.17.10.2	DNS	75	Standard query 0x36a2 A ssl.gstatic.com
161	14.634356	192.168.2.2	192.168.1.1	TCP	60	2681 → 80 [RST] Seq=1 Win=0 Len=0
162	14.634601	192.168.2.2	192.168.1.1	TCP	60	2682 → 80 [RST] Seq=1 Win=0 Len=0
163	14.634840	192.168.2.2	192.168.1.1	TCP	60	2683 → 80 [RST] Seq=1 Win=0 Len=0
164	14.635074	192.168.2.2	192.168.1.1	TCP	60	2684 → 80 [RST] Seq=1 Win=0 Len=0
165	14.635311	192.168.2.2	192.168.1.1	TCP	60	2685 → 80 [RST] Seq=1 Win=0 Len=0
166	14.635546	192.168.2.2	192.168.1.1	TCP	60	2686 → 80 [RST] Seq=1 Win=0 Len=0
167	14.635814	192.168.2.2	192.168.1.1	TCP	60	2687 → 80 [RST] Seq=1 Win=0 Len=0
168	14.636055	192.168.2.2	192.168.1.1	TCP	60	2688 → 80 [RST] Seq=1 Win=0 Len=0
169	14.636291	192.168.2.2	192.168.1.1	TCP	60	2689 → 80 [RST] Seq=1 Win=0 Len=0
170	14.636524	192.168.2.2	192.168.1.1	TCP	60	2690 → 80 [RST] Seq=1 Win=0 Len=0
171	14.636758	192.168.2.2	192.168.1.1	TCP	60	2691 → 80 [RST] Seq=1 Win=0 Len=0
172	14.636993	192.168.2.2	192.168.1.1	TCP	60	2692 → 80 [RST] Seq=1 Win=0 Len=0
173	14.637224	192.168.2.2	192.168.1.1	TCP	60	2693 → 80 [RST] Seq=1 Win=0 Len=0
174	14.637457	192.168.2.2	192.168.1.1	TCP	60	2694 → 80 [RST] Seq=1 Win=0 Len=0
175	14.637692	192.168.2.2	192.168.1.1	TCP	60	2695 → 80 [RST] Seq=1 Win=0 Len=0
176	14.637955	192.168.2.2	192.168.1.1	TCP	60	2696 → 80 [RST] Seq=1 Win=0 Len=0
177	14.638171	192.168.2.2	192.168.1.1	TCP	60	2697 → 80 [RST] Seq=1 Win=0 Len=0
178	14.638405	192.168.2.2	192.168.1.1	TCP	60	2698 → 80 [RST] Seq=1 Win=0 Len=0
179	14.998529	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.2? Tell 172.17.13.13
180	15.468667	192.168.1.2	172.17.10.2	DNS	79	Standard query 0xd650 HTTPS clients4.google.com
181	15.469091	192.168.1.2	172.17.10.2	DNS	79	Standard query 0xd5d1 A clients4.google.com

Pour un flooding (le routeur n'en a laissé passé aucune) :

No.	Time	Source	Destination	Protocol	Lengtl	Info
79	8.055744	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)
80	8.509204	192.168.1.2	172.17.10.1	DNS	75	Standard query 0xc7be HTTPS ssl.gstatic.com
81	8.509612	192.168.1.2	172.17.10.1	DNS	75	Standard query 0x7f7a A ssl.gstatic.com
82	8.592472	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)
83	8.733953	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.0.254? Tell 172.17.13.13
84	8.733986	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.1? Tell 172.17.13.13
85	9.053530	192.168.1.2	172.17.10.1	DNS	75	Standard query 0xa65a A play.google.com
86	9.053776	192.168.1.2	172.17.10.2	DNS	75	Standard query 0xa65a A play.google.com
87	9.119898	192.168.1.2	172.17.10.1	DNS	79	Standard query 0x4c3a HTTPS clients4.google.com
88	9.120128	192.168.1.2	172.17.10.1	DNS	79	Standard query 0x41ce A clients4.google.com
89	9.131429	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)
90	9.217718	192.168.1.2	172.17.10.1	DNS	74	Standard query 0x064f A www.google.com
91	9.217718	192.168.1.2	172.17.10.2	DNS	74	Standard query 0x064f A www.google.com
92	9.520278	192.168.1.2	172.17.10.2	DNS	75	Standard query 0xb9fe HTTPS ssl.gstatic.com
93	9.520576	192.168.1.2	172.17.10.2	DNS	75	Standard query 0x00ed A ssl.gstatic.com
94	9.566513	192.168.1.2	172.17.10.2	DNS	75	Standard query 0xaf06 HTTPS play.google.com
95	9.566729	192.168.1.2	172.17.10.2	DNS	75	Standard query 0xd7a2 A play.google.com
96	9.589966	Cisco_09:36:cd	Cisco_09:36:cd	LOOP	60	Reply
97	9.609097	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.2? Tell 172.17.13.13
98	9.687864	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)
99	9.733876	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.0.254? Tell 172.17.13.13
100	9.733890	GigaByteTech_e6:4d:…	Broadcast	ARP	42	Who has 172.17.10.1? Tell 172.17.13.13
101	10.132011	192.168.1.2	172.17.10.2	DNS	79	Standard query 0x9573 HTTPS clients4.google.com
102	10.132289	192.168.1.2	172.17.10.2	DNS	79	Standard query 0x0856 A clients4.google.com
103	10.222682	192.168.1.254	192.168.1.2	ICMP	70	Destination unreachable (Host unreachable)